

Hand-visible analytic derivations for 103 finite-ledger predicates

1 Scope and logical convention

This note exposes, in hand-checkable form, the predicates not assigned to the other five ledger packets. They are precisely

11 threshold predicates + 48 zero predicates + 43 lower-closure predicates + 1 cap-74 predicate = 103.

The omitted lower-closure predicate is the single six-cap tuple identity; it belongs to the cap-inventory packet. Every inequality below is proved from a displayed analytic inequality, integer arithmetic, rational cross multiplication, or positive-side squaring. An executable ledger may check the same arithmetic, but no executable output is used as a premise.

We use

$$\omega_0 = \frac{\sqrt{3}}{2\pi} - \frac{1}{6}, \quad C_0 = 1 + \frac{8\sqrt{2}}{15}, \quad \rho_* = \frac{\omega_0}{2C_0}, \quad \rho_0 = \frac{1}{\sqrt{337}}, \quad \sigma = \frac{3\omega_0}{4}, \quad \rho_c = \frac{1}{1+2\pi}, \quad d = \frac{\sqrt{337}}{2}.$$

The elementary enclosure

$$\frac{333}{106} < \pi < \frac{355}{113} < \frac{22}{7} \tag{1}$$

is analytic: the lower and middle bounds follow from alternating truncations of Machin's identity $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$, and the last one is immediate by cross multiplication. For reference, 355/113 exceeds the upper Machin truncation by the positive rational numerator 45167474711 over denominator 189820334689453125. The classical integral

$$0 < \int_0^1 \frac{x^4(1-x)^4}{1+x^2} dx = \frac{22}{7} - \pi$$

also gives the final upper bound directly.

2 The eleven threshold predicates

First we obtain a rational two-sided enclosure for ω_0 . The exact integer residuals

$$3 \, 1101^2 113^2 - 607^2 355^2 = 1998482 > 0, \quad 25 \, 333^2 - 243 \, 106^2 = 41877 > 0$$

and (1) imply

$$\sqrt{3} > \frac{607 \, 355}{1101 \, 113} > \frac{607\pi}{1101}, \quad 9\sqrt{3} < 5\pi.$$

Consequently

$$\frac{40}{367} < \omega_0 < \frac{1}{9}. \tag{2}$$

Indeed, $607/2202 - 1/6 = 40/367$, while $5/18 - 1/6 = 1/9$.

Since $2 \cdot 32^2 - 45^2 = 23 > 0$, one has $\sqrt{2} > 45/32$, hence $C_0 > 7/4$ and $\rho_* < 2/63$. The following chain consists entirely of strict rational comparisons, with the indicated radical comparisons justified by squaring:

$$\rho_* < \frac{2}{63} < \frac{1}{\sqrt{337}} < \frac{1}{18} < \frac{3}{40} < \frac{3\omega_0}{4} < \frac{1}{12} < \frac{7}{51} < \frac{1}{1+2\pi} < \frac{1}{7} < \frac{39}{50} < \frac{7}{8}.$$

Here

$$4 \cdot 337 < 63^2, \text{quad}18^2 < 337, \text{quad}367 < 400, \quad 51 < 84, \text{quad}367 < 420,$$

and $7/51 < \rho_c < 1/7$ follows respectively from $\pi < 22/7$ and $\pi > 3$. This proves the five ratio-order rows. It also proves

$$\omega_0 < \frac{1}{9} < \frac{7}{51} < \rho_c.$$

Moreover

$$\omega_0 d > \frac{20\sqrt{337}}{367} > 1, \quad 400 \cdot 337 - 367^2 = 111 > 0.$$

The four terminal comparisons follow from the same bounds:

$$\begin{aligned} \sqrt{114} &< 18 < \frac{2}{\omega_0}, & \frac{2}{\omega_0} &< \frac{367}{20}, \\ \rho_c &< \frac{1}{7} < \frac{60}{367} < \frac{3\omega_0}{2}, & \omega_0 \sqrt{114} &> \frac{40\sqrt{114}}{367} > 1. \end{aligned}$$

The last positive-side squaring has residual

$$1600 \cdot 114 - 367^2 = 47711 > 0.$$

Thus the one-to-one threshold registry is

ID	source row	proved predicate
T1–T5	1–5	$\rho_* < \rho_0 < \sigma < \rho_c < 39/50 < 7/8$, one adjacent pair per row
T6	6	$\omega_0 < \rho_c$
T7	7	$\omega_0 d > 1$
T8	31	$\sqrt{114} < 2/\omega_0$
T9	32	$2/\omega_0 < 367/20$
T10	33	$\rho_c < 3\omega_0/2$
T11	34	$\omega_0 \sqrt{114} > 1$

3 The forty-eight zero specializations

3.1 A recurrence and elementary tangent bounds

Put

$$j_\ell(x) = \sqrt{\frac{\pi}{2x}} J_{\ell+1/2}(x), \quad P_\ell(x) = x^{\ell+1} j_\ell(x).$$

Then

$$P_{\ell+1} = (2\ell+1)P_\ell - x^2 P_{\ell-1}, \quad P_0 = \sin x, \text{quad} P_1 = \sin x - x \cos x. \quad (3)$$

Write $P_\ell = A_\ell(t) \sin x + x C_\ell(t) \cos x$, $t = x^2$. The recurrence gives

$$\begin{aligned} A_2 &= 3 - t, & C_2 &= -3, \\ A_3 &= 15 - 6t, & C_3 &= t - 15, \\ A_5 &= 945 - 420t + 15t^2, & C_5 &= -945 + 105t - t^2, \\ A_6 &= 10395 - 4725t + 210t^2 - t^3, & C_6 &= -10395 + 1260t - 21t^2, \\ A_7 &= 135135 - 62370t + 3150t^2 - 28t^3, & C_7 &= -135135 + 17325t - 378t^2 + t^3. \end{aligned}$$

For $0 < s < \pi/2$, differentiation and integration give

$$\tan s > s, \text{quad} \tan s > s + \frac{s^3}{3}. \quad (4)$$

For $0 < s < 1$, $\sin s < s$ and $\cos s > 1 - s^2/2$ give

$$\tan s < \frac{s}{1 - s^2/2}. \quad (5)$$

3.2 Sixteen half-period locations

For each internal face, the first displayed margin below is $w - (h_L/2)(22/7)$; the second is $(h_R/2)(333/106) - w$. Positivity therefore proves both strict half-period location rows without a decimal approximation.

(ℓ, n)	w	(h_L, h_R)	left margin	right margin
(1, 2)	77/10	(4, 5)	99/70	163/1060
(2, 2)	9	(5, 6)	8/7	45/106
(3, 2)	103/10	(6, 7)	61/70	737/1060
(5, 2)	129/10	(8, 9)	23/70	1311/1060
(1, 3)	21/2	(6, 7)	15/14	105/212
(2, 3)	61/5	(7, 8)	6/5	97/265
(6, 1)	10	(6, 7)	4/7	211/212
(7, 1)	23/2	(7, 8)	1/2	113/106

3.3 Sixteen half-period signs

At an integral or half-integral multiple of π , exactly one of $\sin x, \cos x$ vanishes. Direct substitution in the displayed polynomials gives the following complete sign table.

(ℓ, n)	h_L	left sign	(ℓ, n)	h_R	right sign
(1, 2)	4	—	(1, 2)	5	+
(2, 2)	5	—	(2, 2)	6	+
(3, 2)	6	—	(3, 2)	7	+
(5, 2)	8	—	(5, 2)	9	+
(1, 3)	6	+	(1, 3)	7	—
(2, 3)	7	+	(2, 3)	8	—
(6, 1)	6	+	(6, 1)	7	—
(7, 1)	7	+	(7, 1)	8	—

Here are explicit polynomial witnesses for the only signs not immediate from $A_1 = 1, C_1 = -1, A_2 = 3 - t, C_2 = -3, A_3 = 15 - 6t, C_3 = t - 15$. From (1), at $x = 4\pi$ one has $144 < t < 159$, at $x = 9\pi/2$ one

has $t > 182$, and at $x = 7\pi/2$ one has $110 < t < 121$. Monotonicity on these rational intervals and direct evaluation give

$$C_5(t) < C_5(144) = -6561, \text{quad} A_5(t) > A_5(182) = 421365,$$

$$C_6(t) < C_6(81) = -46116, \text{quad} A_6(t) > A_6(110) = 700645,$$

$$A_7(t) < A_7(110) = -5878565, \text{quad} C_7(t) < C_7(144) = -2492559.$$

For example, $A'_6 = -4725 + 420t - 3t^2$ is positive on $[110, 121]$, while A'_7 is negative there; the other asserted monotonicities follow at once from their displayed derivatives. Combining these signs with the quadrant signs of sine and cosine proves all sixteen entries.

3.4 Eight rational-wall signs

Substitution in (3) reduces each rational-wall sign to one of the following exact tangent comparisons. The shifted angle is always in $(0, \pi/2)$, as already proved by the location table.

(ℓ, n)	wall	hand certificate	$\text{sgn } j_\ell(w)$
(1, 2)	77/10	For $u = 5\pi/2 - w$, $\tan u > 3/20 > 10/77 = 1/w$; hence $\tan(w - 2\pi) = \cot u < w$.	−
(2, 2)	9	For $u = 3\pi - w$, $\tan u > 21/50 > 9/26 = 3w/(w^2 - 3)$.	−
(3, 2)	103/10	For $u = 7\pi/2 - w$, $\tan u > 69/100 > 621540/938227 = (6w^2 - 15)/(w(w^2 - 15))$.	−
(5, 2)	129/10	For $y = w - 4\pi$, $0 < y < 17/50$, so $\tan y < 1700/4711 < 177800829/427700150 = -wC_5/A_5$.	−
(1, 3)	21/2	For $u = 7\pi/2 - w$, $\tan u > 49/100 > 2/21 = 1/w$, hence $\tan(w - 3\pi) < w$.	+
(2, 3)	61/5	For $u = 4\pi - w$, $\tan u > 9/25 > 915/3646 = 3w/(w^2 - 3)$.	+
(6, 1)	10	For $y = w - 3\pi$, $4/7 < y < 29/50$, so $\tan y < 2900/4159 < 188790/127579 = -wC_6/A_6$.	+
(7, 1)	23/2	For $u = 4\pi - w$, $\tan u > u + u^3/3 > 546377/375000 > 3153151489/2276518664 = wC_7/A_7$.	+

For complete cross-multiplication visibility, the four less immediate positive gaps in that table are

$$\begin{aligned} \frac{69}{100} - \frac{621540}{938227} &= \frac{2583663}{93822700}, \\ \frac{177800829}{427700150} - \frac{1700}{4711} &= \frac{110529450419}{2014895406650}, \\ \frac{188790}{127579} - \frac{2900}{4159} &= \frac{415198510}{530601061}, \\ \frac{546377}{375000} - \frac{3153151489}{2276518664} &= \frac{3837851856583}{53355906187500}. \end{aligned}$$

The four small gaps are respectively $31/1540, 24/325, 829/2100, 9939/91150$.

3.5 Six Lorch rows and two wall-order rows

For $\nu = \ell + 1/2$, Lorch's strict analytic first-zero estimate is

$$j_{\nu,1}^2 > \frac{24(\nu + 1)^2}{1 - 2\nu + \sqrt{(2\nu + 3)(2\nu + 11)}} - 2(\nu^2 - 1) =: R_\ell.$$

After rationalization,

$$R_\ell = \frac{3(2\ell + 3)\sqrt{(\ell + 2)(\ell + 6)} - 2\ell^2 + \ell + 6}{4}.$$

All sides being squared are positive, and the following integer residuals prove the three required strict specializations:

ℓ	wall c	positive squared residual proving $R_\ell > c^2$
8	71/6	$1026^2 \cdot 35 - 6067^2 = 35171$
9	64/5	$1575^2 \cdot 165 - 20059^2 = 6939644$
10	69/5	$3450^2 \cdot 3 - 5911^2 = 767579$

Since $j_\ell(x) > 0$ for sufficiently small $x > 0$, and its first positive zero is strictly to the right of c , these same three rows imply $j_\ell(c) > 0$. Thus the three executable sign rows are analytic corollaries, not independent numerical assumptions. Finally,

$$\frac{103}{10} - \frac{81}{8} = \frac{7}{40} > 0, \quad \frac{61}{5} - \frac{21}{2} = \frac{17}{10} > 0.$$

The one-to-one zero registry is as follows. Number the eight internal faces in the order of the location table. For face r , IDs $Z_{5r-4}, \dots, Z_{5r}$ are, in order: right of the left half-period, left of the right half-period, left half-period sign, right half-period sign, and rational-wall sign. This bijects IDs $Z1$ – $Z40$ with the first forty source rows. The remaining bijection is

ID	source row	predicate
Z41	41	Lorch specialization $\ell = 8, c = 71/6$
Z42	42	positive sign at $\ell = 8, c = 71/6$
Z43	43	Lorch specialization $\ell = 9, c = 64/5$
Z44	44	positive sign at $\ell = 9, c = 64/5$
Z45	45	Lorch specialization $\ell = 10, c = 69/5$
Z46	46	positive sign at $\ell = 10, c = 69/5$
Z47	47	$103/10 > 81/8$
Z48	48	$61/5 > 21/2$

4 The forty-three lower-closure predicates

4.1 Eight activation and exclusion rows

The complete arithmetic is

$$\begin{aligned}
(23/2)^2 - 114 &= 73/4 > 0, & 10^2 &= 100, \\
(129/10)^2 - 114 &= 5241/100 > 0, & 103/10 - 81/8 &= 7/40 > 0, \\
(81/8)^2 + 8 &= 7073/64, & (81/8)^2 + 18 - 114 &= 417/64 > 0, \\
(21/2)^2 + 4 - 114 &= 1/4 > 0, & 16\pi^2 &> 16 \cdot 3^2 > 114.
\end{aligned}$$

These are exactly lower IDs L1–L8, corresponding to source rows 1–8.

4.2 Five Weyl payments

For $0 < \rho < \rho_c < 1/7$, (1) yields the uniform analytic lower bound

$$W(\rho, K) = \frac{2}{9\pi}(1 - \rho^3)K^3 > \frac{38}{539}K^3. \quad (6)$$

Using $d > 9$ and $\sqrt{7073}/8 > 21/2$, the five payments are the following exact positive reserves:

ID	source row	threshold/cap	reserve from (6)
L9	10	$K > d, 46$	$(38/539)9^3 - 46 = 2908/539$
L10	11	$K > 10, 59$	$(38/539)10^3 - 59 = 6199/539$
L11	12	$K > 81/8, 66$	$(38/539)(81/8)^3 - 66 = 990435/137984$
L12	13	$K > 21/2, 69$	$(38/539)(21/2)^3 - 69 = 555/44$
L13	14	$K > \sqrt{7073}/8, 78$	$(38/539)(21/2)^3 - 78 = 159/44$

4.3 The cap-395 cell: eighteen rows

Put $R = 367/20$, $z_\ell = \ell + 1/2$. The analytic turning estimate

$$G_K(z) < \frac{(K - z)^{3/2}}{3\sqrt{K}}$$

is increasing in K , so on $K < R$ a strict cap c_ℓ follows from

$$\frac{(R - z_\ell)^3}{9R} = \frac{(357 - 20\ell)^3}{1321200} < \left(c_\ell + \frac{3}{4}\right)^2. \quad (7)$$

The following table gives all fifteen caps and the exact positive numerator $82575(4c_\ell + 3)^2 - (357 - 20\ell)^3$ obtained by clearing the positive denominators in (7).

ℓ	0	1	2	3	4	5	6	7
c_ℓ	6	5	5	4	4	3	3	3
margin	14697882	5409422	11827162	3611502	8555642	1604782	5267322	8361062

ℓ	8	9	10	11	12	13	14
c_ℓ	2	2	1	1	1	1	0
margin	2346202	4446342	176282	1474822	2444562	3133502	286642

Thus L15–L29 correspond bijectively to source rows 16–30, with L($15 + \ell$) the ℓ -th table column. Since $G_R(z)$ is strictly decreasing in z , the $c_{14} = 0$ row removes every $\ell \geq 14$ tail; this is L30/source row 31.

The weighted cap itself, L14/source row 15, is visible from the contributions

$$6, 15, 25, 28, 36, 33, 39, 45, 34, 38, 21, 23, 25, 27, \quad \sum_{\ell=0}^{13} (2\ell + 1)c_\ell = 395.$$

Finally, on the exceptional cell $\rho K \geq 5/2$, with $\rho < 11/80$ and $1/\pi > 113/355$,

$$\begin{aligned} W(\rho, K) &> \frac{2}{9} \frac{113}{355} \left(\frac{5}{2}\right)^3 \left[\left(\frac{80}{11}\right)^3 - 1 \right] \\ &= \frac{160293325}{378004} = 395 + \frac{10981745}{378004} > 395. \end{aligned}$$

This is L31/source row 32.

4.4 Twelve floor and boundary-owner rows

The nine finite arithmetic rows are instances of one identity:

$$\frac{3}{2}(p+1) \leq 2p + \frac{1}{2} \quad (p \geq 2), \text{quad} \left(2p + \frac{1}{2}\right) - \frac{3}{2}(p+1) = \frac{p-2}{2} \geq 0. \quad (8)$$

At $p = 2$, $(3/2)3 = 9/2$; this is L32/source row 33. For $p = 2, \dots, 9$, formula (8) is respectively L33–L40/source rows 34–41.

The half-integer face is exact: $5/2 = 2 + 1/2$. Assigning it to the high side makes the interface remainder vanish and leaves the exceptional cell closed; this is L41/source row 42. At $K = 2/\omega_0$,

$$p = \lfloor \omega_0 K \rfloor = 2, \quad \rho K \leq \frac{2\rho_c}{\omega_0} < 3$$

by T10. This proves L42/source row 43. Moreover

$$\rho K - \frac{1}{2} < \frac{5}{2}, \quad \text{quad} m = \left\lfloor \rho K - \frac{1}{2} \right\rfloor \leq 2 \leq 2p - 1 = 3,$$

and the strict exceptional condition $K < 2/\omega_0$ fails at equality. Thus the equality face is aggregate-owned, proving L43/source row 44. More generally, if $K \geq 2/\omega_0$, then $p \geq 2$ and

$$\rho K < \frac{3}{2}(p + 1) \leq 2p + \frac{1}{2}, \quad m \leq 2p - 1,$$

so the finite instances conceal no unproved large- p case.

5 The supplemental cap–74 incompatibility

The propagated ball wall requires $z^2 > 2409/100$, while the product wall requires $z^2 < 70/3$. They are incompatible because

$$\frac{2409}{100} - \frac{70}{3} = \frac{7227 - 7000}{300} = \frac{227}{300} > 0.$$

This is supplemental ID S1 and is a hand proof, not merely a reported certificate value.

6 Census and source-visibility verdict

family	count	IDs	hand-visible status before this note
threshold remainder	11	T1–T11	printed, with a few implications compressed
zero specializations	48	Z1–Z48	tables printed; several recurrence steps compressed
lower remainder	43	L1–L43	payments printed; floor instances generated implicitly
supplemental cap–74	1	S1	printed exactly
total	103		

No predicate in this 103-row allocation is an unsupported executable-only mathematical premise. Before this note, the principal visibility gaps were the one-to-one expansion of the forty zero sign/location rows, the fact that the three Lorch-wall signs are logical corollaries of the strict first-zero bounds, and the expansion of the twelve floor/boundary rows. The displayed recurrences, rational margins, general floor identity, and exact ownership arguments close those presentation gaps.